

Grant DeWaay

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EDUCATION

Iowa State University (Cumulative GPA: 3.3)

Fall 2020 – Exp. Fall 2024

Major: B.S. Software Engineering

Secondary Major: Interdisciplinary Design – Area of study focused on design thinking, project management, leadership, and product development.

RELEVANT DEVELOPER SKILLS

Programming Languages: Java, Python, C#, Golang, Java/TypeScript, C, HTML/CSS/JS

Libraries and Frameworks: React, Next.js, Spring Boot, .NET Core, Express, Flask

Database Technologies: MySQL, MongoDB, Microsoft SQL Server, Neo4j, Chroma DB

Project Management: GitHub / GitLab, Azure DevOps, Figma, Trello

Deployment: Docker, AWS

WORK EXPERIENCE

Buildertrend – Summer Software Engineering Internship

Summer 2023 & 2024

AI-Driven Rebate Redemption Automation and Product Matching – Summer 2024

Technologies used: C#, .NET Core, React, TypeScript, Artificial Intelligence, MongoDB, OpenAI, Microsoft SQL Server

- **Enhanced rebate matching accuracy by 10%** by implementing a GPT-4o LLM pipeline that improved AI-driven product searches for rebate-eligible items within RebateHawk, streamlining refunds for Buildertrend's users.
- **Designed a scalable architecture for flexible LLM integration**, enabling seamless switching between OpenAI and Google Gemini, accelerating model experimentation and boosting system adaptability.
- **Reduced manual validation time for testers** by implementing semantic search confidence filtering, optimizing workflows and increasing efficiency.
- **Collaborated in an Agile team** using Microsoft Azure for CI/CD and project management, ensuring secure deployments, high-quality code, and reliable project rollouts.

Semantic Search Prototype – Summer 2023

Technologies used: Python, React, TypeScript, Artificial Intelligence, OpenAI, Chroma DB, LangChain

- **Improved search accuracy by 30%** by developing a semantic search prototype with OpenAI's embedding models, providing a scalable foundation for future AI-driven search solutions.
- **Increased interest in AI-driven search** by designing a React-based testing environment, enabling Buildertrend to explore and expand on the prototype on a team I would join the following summer.
- **Enabled cross-team integration** by building a Flask backend to serialize and query search results, facilitating seamless adoption of the semantic search engine across applications.

PROJECTS

Optimal Play AI Checkers Bot

Technologies used: Java, Monte Carlo Tree Search (MCTS), AI-based Game Theory, GUI Integration, Adversarial Search

- **Developed an MCTS-based AI** that analyzes game states to suggest optimal moves, implementing selection, expansion, simulation, and backpropagation steps with the UCB1 formula and a $\sqrt{2}$ exploration constant for balance.
- **Designed code for move generation** that enforces all game rules, including mandatory jumps and multi-jump sequences.
- **Demonstrated advanced AI game mechanics**, applying MCTS in an adversarial game context
- **Developed a GUI** to visually indicate valid moves, selections, and jump sequences, with user-friendly prompts and real-time feedback.
- **Conducted performance testing** on different exploration constants to evaluate AI move quality and response, documenting the findings.

Code42 – Programming Learning Platform for Android Devices

Technologies used: Android Studio, Java, Spring Boot, Python, Go, MySQL

- **Developed an Android Application to teach kids how to code** in various languages, including Java, Python, and Go. Similar to platforms like Zybooks and LeetCode but with class management.
- **Custom coding assignments** can be entered by professors, where they can assign points, the language, test cases, and a description
- **Classroom management:** Teachers can create new courses, add students, and change grades for assignments.
- **Code is executed safely** on our servers, with multiple methods of safe code execution used, ensuring that no malicious code is permitted.

AWARDS

HackHarvard 2024 Hackathon | Funniest Hack Award

October 2024

Iowa State College-by-College Pitch Off | 1st Place – New Idea

February 2024

Student Innovation Fund Challenge | 1st Place

May 2023

Game2Work Social Impact Game Jam | Critic's Choice Award

February 2021